Michael Chan

chunheimichael.chan@epfl.ch | miki998.github.io | github.com/miki998 | linkedin | Lausanne, Switzerland

Profile

EE PhD candidate at EPFL working on **directed graph signal processing and graph neural networks**: spectral methods, non-parametric null models, and statistical inference on high-dimensional, noisy time series. Four industry internships (Intuitive Surgical, Logitech, Besedo, SGI) shipping ML pipelines that recover weak signals from noisy, non-stationary data. Seeking quantitative research / ML roles where signal extraction, rigorous statistical validation and modelling are the core problem.

Core strengths: time-series & spectral analysis ; hypothesis testing, surrogate & bootstrap methods ; graph neural networks ; stochastic processes ; publishing at top-tier venues

Education

PhD, Electrical Engineering | Swiss Federal Institute of Technology Lausanne (EPFL) — Oct 2023 – Present
Thesis: "Directed Graph Signal Processing and applications in Neuroimaging" | Medical Image Processing Lab (MIP:Lab), Neuro-X

M.Sc., Data Science | EPFL — Sep 2020 – Jul 2023

B.Sc., Mathematics | École Polytechnique, Palaiseau, France — Sep 2017 – Jul 2020

Technical Skills

Programming: Python, C++, C, SQL, Bash/Unix, COQ

ML & scientific stack: PyTorch, scikit-learn, NumPy/SciPy, pandas, NetworkX, CUDA/multi-GPU training, Jupyter

Data & infrastructure: Git, Docker, PySpark, LaTeX, HPC/cluster workflows

Methods: graph neural networks, transformers, RNNs/sequence models, spectral & time-series analysis, statistical hypothesis testing, surrogate-data & bootstrap inference, dimensionality reduction, numerical optimization, adversarial robustness, object detection & segmentation

Open source: maintainer of **NiGSP** (graph signal processing package for neuroimaging)

Industry Experience

Algorithm Team Intern | Intuitive Surgical Inc. — Jul 2022 – Sep 2022

- Built an end-to-end pipeline recovering **sub-percent-amplitude pulsatile signals** from endoscopic video streams (rPPG via plane-orthogonal-to-skin), using lock-in amplitude detection as a tissue-perfusion proxy.
- Raised map SNR by tracking and warping image patches across frames to suppress motion-induced noise, improving amplitude-estimate stability.
- Delivered a proof-of-concept Amplitude Pulsatile Component mapping tool adopted by the algorithm team for real-time surgical video analysis.

CTO AI Team Intern | Logitech International S.A. — Sep 2021 – Feb 2022

- Designed controlled experiments isolating hardware vs. modelling contributions to **heart-rate and HRV estimation accuracy**, reducing error under motion and low-light conditions.
- Showed high-frame-rate and infra-red-sensitive (from camera) improves robustness in adversarial lighting, proving the value of multi-spectral imaging for webcams.
- Developed attention-based and 3D-CNN models for rPPG, with interpretability analysis identifying the spatial regions and frequency bands driving predictions.

Computer Vision Research Intern | Besedo Content Moderation — Jun 2020 – Sep 2020

- Shipped a **high-precision watermark detector** that automatically filters watermarked user uploads on retail marketplaces, tuned at a fixed-recall operating point to keep false removals near zero.
- Combined transfer learning on a general object-detection backbone with a downstream statistical model.

Data Science Intern | SGI Venture Limited — Jun 2019 – Aug 2019

- Built pet facial recognition under severe **small-sample constraints** (YOLO detection + LBPH feature matching).

Research Experience

Doctoral Researcher | MIP:Lab, EPFL Oct 2023 – Present

- Developed novel graph neural networks, introducing **TwinS-GCN** featuring **non-holomorphic spectral filters** and leveraging functional calculus, that improves performance on directed graphs.
- Defined a **graph diffusion–advection operator** leveraging partial differential equations and physically motivated principles to extends spectral analysis to directed networks.
- Built theory for **non-parametric surrogate-data generators** for significance testing on graphs — null models that preserve nuisance structure while destroying the effect under test, the graph analogue of block-bootstrap and permutation testing.
- Extended the **Hilbert transform to graph signals**, defining instantaneous phase and analytic representations on networks.
- Co-developed **bimodularity**, a directed-network community-detection criterion.
- Applications in neuroimaging for **studies of emotions, and brain connectivity** in patients with neurological disorders.
- **Output:** 10 journal publications, 4 further submissions under review, 4 conference papers, 2 talks and 5 posters.

Selected Publications

- "TwinS-GCN: Spectral Conjugate for Spectral Graph Convolutional Networks" — under submission, ICLR 2027
- "Graph Diffusion-Advection Operator for Directed Graph Signal Processing" — in review, IEEE TSIPN (2026)
- "Statistical Testing on Directed Graphs by Surrogate Data Generation" — IEEE TSIPN (2026)
- "Hilbert Transform on Graphs: Let There Be Phase" — IEEE SPL (2025)
- "Community Detection for Directed Networks Revisited Using Bimodularity" — PNAS (2025)

Further work in Science Advances, Nature Communications, Human Brain Mapping, Network Neuroscience, Imaging Neuroscience; full list at miki998.github.io.

Awards & Competitions

- 2nd Runner-up, Data Analytics Competition — Facebook Analytics, 2020
- 1st Runner-up, Research Student Award — Swiss Society for Bioengineering (SSBE), 2023
- Conference Travel Grant — ICASSP 2026
- Scholarship Holder, Undergraduate Studies — Bourse d'excellence Campus France
- Regional phase, HULT Prize (social entrepreneurship)

Leadership & Teaching

- **Head Teaching Assistant**, Neural Signals and Signal Processing (NX-421), EPFL 2023 – Present
  Manage a team of **7–10** teaching assistants for course and exam logistics for **170+** students.
- **Project supervision:** mentored **12** Bachelor/Master students through independent research projects, from scoping to written deliverable.
- **Teaching Assistant**, Image Processing I/II (MICRO-511/512), EPFL 2023 – Present

Additional

Languages: English (fluent), French (native), Cantonese/Mandarin (native), Spanish (intermediate), Japanese (intermediate)
Interests: ABRSM Grade 8 piano; Hapkido (blue belt)/Martial Arts; table tennis (3rd place, Toamasina community championship); volunteer event host and translator for cultural festivals.